

MOUNT GELLIBRAND, AustraliaWMO: **958450**Lat: **38.2333S**Lon: **143.7925E**Elev: **262**StdP: **98.22**Time Zone: **10.00 (AUS)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	4.0	-0.1	3.9	16.4	1.1	4.2	11.4	20.4	9.2	18.9	9.7	7.0	280	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	33.6	17.7	30.7	17.1	27.9	16.7	19.8	26.2	18.9	25.3	18.0	24.6	8.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.6	20.4	17.2	12.7	19.9	16.1	11.8	19.1	57.8	25.8	54.6	25.2	51.9	24.8	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.1	14.1	1.6	39.1	0.7	2.4	1.1	40.8	0.7	42.2	0.3	43.5	-0.1	45.3		
(5)			1.3	21.9	0.7	1.1	0.8	22.7	0.3	23.3	-0.1	24.0	-0.6	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	18.4	18.3	17.0	14.4	11.4	9.1	8.4	8.8	10.1	12.1	14.4	16.3
	DBStd	4.80	4.26	3.77	3.89	3.19	2.40	1.67	1.55	2.03	2.54	3.46	3.92	3.98
	HDD10.0	199	0	0	0	2	11	37	55	49	28	13	3	1
	HDD18.3	2057	51	43	74	124	216	278	309	296	247	198	132	89
	CDD10.0	1366	261	233	217	135	54	9	4	12	31	77	135	197
	CDD18.3	182	53	43	33	7	0	0	0	0	0	4	15	27
	CDH23.3	1969	619	396	316	50	1	0	0	0	2	56	173	358
CDH26.7	846	311	169	125	9	0	0	0	0	0	13	55	164	
Wind	WSAvg	7.1	6.8	6.7	6.7	6.5	7.1	6.8	7.9	7.8	7.5	7.4	6.9	7.0
Precipitation	PrecAvg	520	38	32	31	37	37	54	51	52	60	50	44	36
	PrecMax	749	143	91	79	130	68	114	95	95	126	100	98	125
	PrecMin	299	0	0	9	3	7	12	23	16	26	4	11	1
	PrecStd	102	33	23	17	28	17	21	19	22	25	29	23	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	35.1	34.5	27.9	21.2	15.4	14.6	17.5	21.7	28.5	31.5	35.8
		MCWB	18.4	18.5	17.2	15.5	13.1	11.6	11.4	12.1	13.3	15.3	17.3	17.1
	2%	DB	33.9	31.4	30.2	24.5	18.4	13.7	13.0	15.3	18.7	24.3	28.0	31.5
		MCWB	17.7	17.5	16.6	14.2	12.6	11.1	10.5	10.9	13.1	14.7	16.6	16.8
	5%	DB	30.0	28.3	26.8	22.0	16.6	12.6	11.8	13.5	16.4	21.2	24.8	27.6
		MCWB	17.1	17.7	16.2	13.9	12.2	10.7	9.7	10.2	12.0	13.9	16.2	16.2
	10%	DB	26.1	25.0	23.6	19.5	15.0	11.7	11.0	12.0	14.6	18.1	21.6	23.8
		MCWB	16.8	17.0	15.9	13.6	12.0	10.1	9.2	9.5	11.0	13.0	15.1	15.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.0	19.5	17.1	15.5	13.0	12.7	13.1	15.4	17.6	20.1	20.3
		MCDB	28.8	28.1	25.0	22.5	16.9	13.8	13.5	16.4	18.8	23.7	26.4	24.6
	2%	WB	19.7	19.5	18.3	16.1	14.1	11.9	11.1	11.7	13.9	15.7	18.2	18.5
		MCDB	27.1	25.3	24.5	20.2	16.3	12.9	12.0	13.9	17.2	21.6	24.0	25.5
	5%	WB	18.7	18.6	17.4	15.2	13.1	11.1	10.2	10.7	12.6	14.5	16.9	17.4
		MCDB	26.3	24.3	23.4	18.8	15.6	12.1	11.2	12.6	15.8	19.7	22.7	24.6
	10%	WB	17.7	17.7	16.5	14.4	12.2	10.4	9.5	9.9	11.5	13.4	15.6	16.3
		MCDB	25.0	23.3	22.0	18.0	14.4	11.4	10.5	11.6	14.1	17.4	20.8	23.2
Mean Daily Temperature Range	5% DB	MDBR	12.0	10.9	9.8	8.1	5.9	4.9	5.1	6.0	7.2	9.0	10.0	11.3
		MCDBR	18.4	16.6	15.2	11.9	8.2	5.8	5.9	7.7	10.0	14.4	16.0	18.3
		MCWBR	4.9	4.6	4.5	4.1	4.4	4.0	4.2	4.6	5.8	6.4	6.2	5.4
	5% WB	MCDBR	14.4	13.2	12.1	9.5	6.8	5.1	5.5	6.8	9.0	12.1	13.0	13.8
		MCWBR	5.6	5.1	4.9	4.5	4.5	4.0	4.3	4.7	5.9	6.6	6.3	5.6
Clear-Sky Solar Irradiance	taub		0.340	0.338	0.319	0.312	0.303	0.293	0.294	0.308	0.323	0.328	0.336	0.329
	taud		2.511	2.528	2.583	2.577	2.578	2.596	2.585	2.537	2.486	2.485	2.491	2.535
	Ebn at Noon		993	971	950	894	840	821	842	885	929	969	990	1008
	Edh at Noon		111	105	91	80	69	63	68	82	99	108	113	110
All-Sky Solar Radiation	RadAvg		7.08	6.15	4.81	3.35	2.24	1.76	1.96	2.77	3.95	5.30	6.21	7.01
	RadStd		0.44	0.28	0.27	0.20	0.10	0.13	0.12	0.15	0.24	0.41	0.39	0.45